



iPLUG
User Guide
Network Resilience Plugins

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Table of contents

1	UserGuide	4
1.1	Installation	5
1.1.1	Permission access	5
1.1.2	General Requirements	5
1.1.3	Recommended additional Plugins	5
1.1.4	Specific requirement for → Impact Evaluation	5
1.1.5	Plugin Installation steps	5
1.2	Usage	5
1.3	Hazards Combination	6
1.4	Impact Evaluation	8
1.5	Risk Assessment and Resilience Index	11
1.6	Case Study: 220 kV transmission network	13
1.6.1	System description	13
1.6.2	Step-by-Step Guide to Run the Example	14

1 UserGuide

Three Python-based QGIS plug-ins are developed in the context of the iPlug project to address the network's resilience.

1. **Hazard Combination** → This Plugin creates a new layer providing the probability of occurrence of a combined event from simple events. The user should indicate the type of relation between events: dependent or independent.
2. **Impact Evaluation** → This Plugin determines the impact of each element based on an OPF analysis. The user should introduce one layer for buses and one for power lines. The Attribute Table of each layer should include the required parameters to perform the OPF analysis. Besides, the user can select the type of OPF (AC or DC) and the simulation (24-hour profile or instantaneous). As a result, the Plugin created two new layers, one for buses and one for power lines. Each new layer includes, in the attribute table, the impact of each element in power or energy, depending on the simulation type selected. Finally, the Plugin provides a spreadsheet with the same information.
3. **Risk Assessment and Resilience Index** → This Plugin evaluates power/energy at risk in each element based on the hazards that might affect a system and its vulnerabilities. First, the user should indicate the year's season to perform the analysis. Then, the maximum power line's segment length. Finally, the user should upload a spreadsheet with the system's vulnerabilities.

Please, follow the stages described in this User Guide when using the Plugins, otherwise, the results may not be as expected.

1.1 Installation

1.1.1 Permission access

The Plugins are available in a GitHub repository: https://github.com/mmONTALAP/Resilience_Tool.

1.1.2 General Requirements

- **QGIS $\geq$ 3.0:** Download from [QGIS OfficialSite](#).

1.1.3 Recommended additional Plugins

- **Plugin Reloader:** Download from [Plugin Reloader OfficialSite](#).

1.1.4 Specific requirement for → Impact Evaluation

- **Pandapower $\geq$ 6.4:**
 - Open OSGeo4W shell (packed with QGIS in the start menu)
 - Install the library using “pip”, following the instructions at the [Pandapower OfficialSite](#).

1.1.5 Plugin Installation steps

- Download from the repository the required Plugin.
- Unzip the folder.
- Save the folder in the QGIS’ Active User Profile.
 - Linux:
`C:\Users\USER\AppData\Roaming\QGIS\QGIS3\profiles\default\python\Plugins\ORStools`
 - Windows:
`C:\Users\USER\AppData\Roaming\QGIS\QGIS3\profiles\default\python\Plugins\ORStools`
 - Mac OS:
`Library/Application Support/QGIS/QGIS3/profiles/default/python/Plugins/ORStools`

If not found, the QGIS’ Active User Profile Folder can be accessed through “QGIS” ► “Settings”
► “User Profile” ► “Open Active Profile Folder”

1.2 Usage

- Before using a Plugin, using the Reload Plugin through “Plugin Reloader” is recommended.
- Follow the instructions detailed in Sections 1.3 to 1.5 for each Plugin.

1.3 Hazards Combination

A resilience analysis should consider all potential hazards that might affect the system. Different institutions and organizations provide geolocated information about simple hazards. The user should store all simple hazards in a **QgsTreeLayer** called *Hazards*, otherwise the Python-based QGIS Plugin will not consider them. The Layers included in the QgsTreeLayer should contain at least the following information in its Attribute Table:

1. Field *SUMMER* → summer hazard index.
2. Field *WINTER* → winter hazard index.
3. Field *AUTUMN* → autumn hazard index.
4. Field *SPRING* → spring hazard index.

Simple hazards can simultaneously occur, increasing the vulnerabilities of the system. In that context, the Hazard Combination Plugin results in a new layer that includes the probability of having the combined events in each season of the year.

Figure 1 shows the Dialog Box of this Python-based QGIS Plugin. First, the user should select the type of combination between events: dependent or independent. Second, the user should choose *Event A* and *Event B* among all layers included in the QgsTreeLayer called *Hazards*.

The Python-based QGIS Plugin will unlock the $P(A|B)$ boxes when the user chooses dependent events. The user should introduce how the hazard index of *Event A* varies once *Event B* occurs. Introduce this variation using per unit (p.u.) notation: 1 signifies no variation, 0.5 represents half hazard index, and 2 indicates double hazard index.

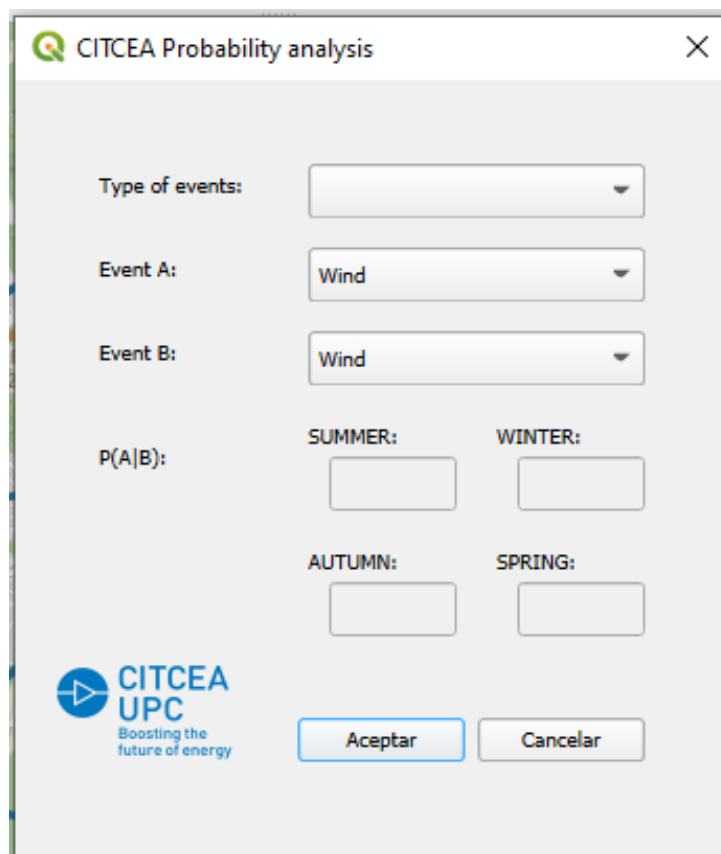


Figure 1: Dialog Box of CITCEA Probability Analysis Python-based QGIS Plugin

The Python-based QGIS Plugin adds a new layer to the hazard database, including the combined hazard index. The new layer is generated as a temporary layer. It should be saved in the project's folder to make it permanent. Afterwards, the layer should be moved to the QgsTreeLayer *Hazards*. Once included, the user can select the layer in subsequent event combinations.

1.4 Impact Evaluation

The Python-based QGIS Plugin, which determines each element's impact, allows the user to determine the amount of energy not supplied when a component, i.e., bus or power line, fails. The impact is determined by comparing the active power supplied when all elements are connected with the active power supplied when the analyzed element fails. The OPF objective function maximizes the active power delivered to loads.

The Python-based QGIS Plugin provides various kinds of analysis. The user can select the year's seasons in which the analysis is desired, the type of OPF, DC or AC, and a daily or instantaneous analysis. The daily analysis considers a 24-hour profile while the instantaneous analysis considers the average power, i.e., the OPF is performed for a single period.

As a result, the Python-based QGIS Plugin generates two new layers, one for buses and one for power lines. Each layer includes information about each element's impact. A .csv file is also created in the folder specified by the user. The .csv includes each element's impact and the total active power supplied by the system when all elements are connected.

A QgsTreeLayer called *System_TBA* should include the layers of the network. At least two layers are requested: one for buses and one for power lines. The Layers included in the QgsTreeLayer should contain at least the following information in the Attribute Table:

1. Buses Layer:

- (a) Field *Name* → bus name
- (b) Field *Voltage_kV* → voltage level
- (c) Field *GRID* → 1 in case of being the 0 connected to the external grid, 0 otherwise.
- (d) Field *Type_GEN* → the generators can be modeled as a PQ or PV. A PQ generator provides constant active and reactive power while a PV provides constant active power while fixing the voltage in a given setpoint.
- (e) Field *PL_Av_Au* → average active power demanded in autumn to the bus in kW.
- (f) Field *PL_Av_Wi* → average active power demanded in winter to the bus in kW.
- (g) Field *PL_Av_Sp* → average active power demanded in spring to the bus in kW.
- (h) Field *PL_Av_Su* → average active power demanded in summer to the bus in kW.
- (i) Field *QL_Av_Au* → average reactive power demanded in autumn to the bus in kW.
- (j) Field *QL_Av_Wi* → average reactive power demanded in winter to the bus in kW.
- (k) Field *QL_Av_Sp* → average reactive power demanded in spring to the bus in kW.
- (l) Field *QL_Av_Su* → average reactive power demanded in summer to the bus in kW.
- (m) Field $\{PL_{00_Au}, \dots, QL_{24_Au}\}^*$ → average active power demanded during the corresponding hour in autumn to the bus in kW (24 fields).

- (n) Field $\{PL_00_Wi, \dots QL_24_Wi\}^*$ → average active power demanded during the corresponding hour in winter to the bus in kW (24 fields).
- (o) Field $\{PL_00_Sp, \dots QL_24_Sp\}^*$ → average active power demanded during the corresponding hour in spring to the bus in kW (24 fields).
- (p) Field $\{PL_00_Su, \dots QL_24_Su\}^*$ → average active power demanded during the corresponding hour in summer to the bus in kW (24 fields).
- (q) Field $\{QL_00_Au, \dots QL_24_Au\}^*$ → average reactive power demanded during the corresponding hour in autumn to the bus in kW (24 fields).
- (r) Field $\{QL_00_Wi, \dots QL_24_Wi\}^*$ → average reactive power demanded during the corresponding hour in winter to the bus in kW (24 fields).
- (s) Field $\{QL_00_Sp, \dots QL_24_Sp\}^*$ → average reactive power demanded during the corresponding hour in spring to the bus in kW (24 fields).
- (t) Field $\{QL_00_Su, \dots QL_24_Su\}^*$ → average reactive power demanded during the corresponding hour in summer to the bus in kW (24 fields).

2. Power Lines Layer:

- (a) Field *Line_Name* → line name
- (b) Field *r_ohm_km* → resistance of the line in Ohm per km
- (c) Field *x_ohm_km* → inductance of the line in Ohm per km
- (d) Field *c_nf_km* → capacitance of the line in nano Fard
- (e) Field *i_max_kA* → maximal thermal current in kA
- (f) Field *PARALLEL* → number of parallel line systems

The *From* and *To* buses, as well as the *Length*, are automatically determined through the geolocated data.

1. * parameters only required in the hourly OPF. They can be omitted in instantaneous OPF.
2. additional fields might included though they are not going to be used in the OPF calculation.

The Dialog Box of the Plugin is represented as Figure 2.

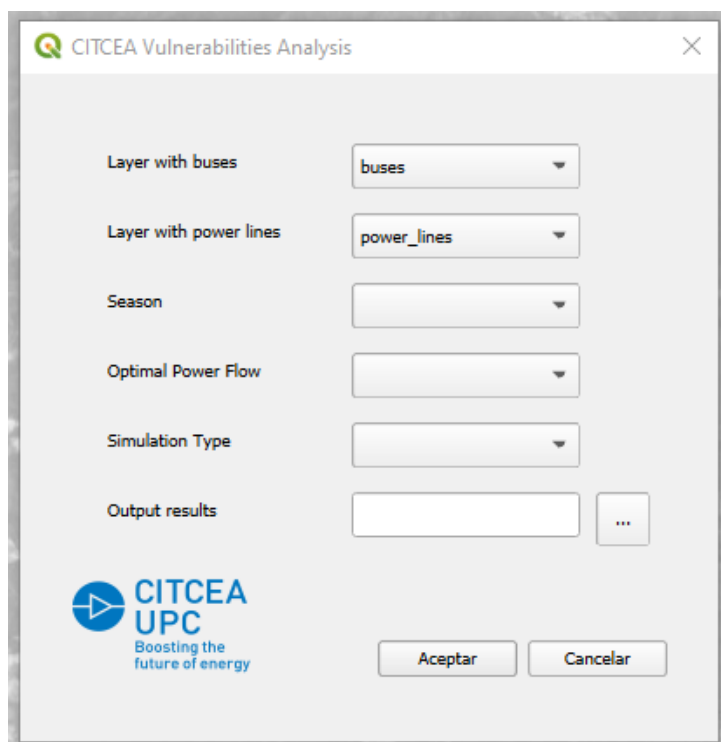


Figure 2: Dialog Box of CITCEA Probability Analysis Python-based QGIS Plugin

First, the user should select the layers with buses and power lines in the first and second dropdown lists, respectively. The list shows all layers included in the QgsTreeLayer *System_TBA*. Second, the user should choose the year's season for the element's impact evaluation. Third, the user can select between a DC and AC OPF depending on the system characteristics, data available, and expected accuracy. Fourth, the user should choose between instantaneous or hourly OPF. Finally, the user should name a .csv file to save the element's impact obtained.

As a result, the Plugin generates two new layers, one for buses and one for power lines. The layers contain the same data in the Attribute Table as the input layers and a new Field called *Impact*. The Field *Impact* includes the amount of energy or power that would be unsupplied in case of losing the element. *Impact* is quantified in terms of power if the user selects an instantaneous analysis. Additionally, the .csv includes a list of elements and their impact. New layers are generated as temporary layers. They should be saved in the project's folder to make them permanent.

1.5 Risk Assessment and Resilience Index

A resilience analysis should consider all hazards affecting a system and its vulnerabilities. The Resilience Analysis Python-based QGIS Plugin enables risk assessment and the calculation of the system's resilience index. Figure 3 shows the Python-based QGIS Plugin Dialog Box.

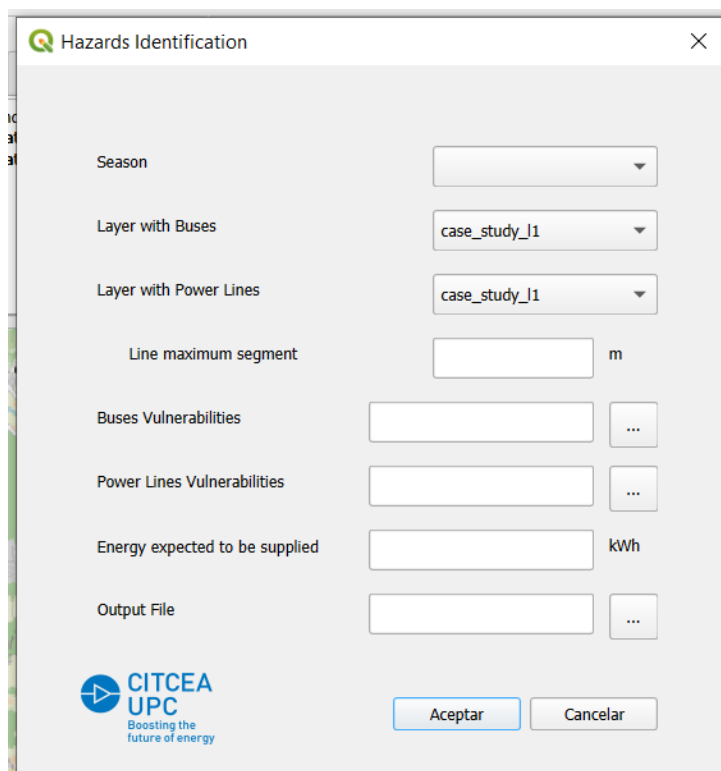


Figure 3: Dialog Box of CITCEA Resilience Index Python-based QGIS Plugin

First, the user should select the year's season to analyze. Second, the user should choose the layer with buses and power lines in the dropdown lists. The dropdown lists include all layers saved in the `QgsTreeLayer System.TBA`. Third, the user can specify the maximum length of each segment of the power lines. In case of not expecting a segmentation in power lines, introduce a value a high value. Fourth, the user should introduce a .csv file with the vulnerabilities of the buses and power lines, respectively. Fifth, the user should manually introduce the sum of total impacts. Finally, the user should specify the .csv file to save the risk assessment results.

Regarding the .csv of the vulnerabilities, the minimum content should be:

1. Column name: *Name* → The name of each bus / power line. The elements should have the same name used in the layers included in `QgsTreeLayer System.TBA`. The name of each element should be repeated for each vulnerability, i.e. if an element has 10 vulnerabilities, the .csv should include 10 rows and each row should include the element's name.
2. Column name: *VULNERABILITY* → vulnerability's name. The user should specify each vulnerability in a different row.

3. Column name: *hazard-name_likelihood* → As many columns as hazards included in the QgsTree-Layer *Hazards* need to be included replacing *hazard-name* by the layer's name. In each row, the user should specify the likelihood of the vulnerability affecting the system once the hazard occurs.
4. Column name: *hazard-name_damage* → As many columns as hazards included in the QgsTreeLayer *Hazards* need to be included replacing *hazard-name* by the layer's name. In each row, the user should specify the impact that the element has on the system. Those impacts can be estimated through the Plugin detailed in Section 1.4.

This Python-based QGIS Plugin adds two more layers to the QGIS project for buses and power lines. The Attribute Table of each layer includes the name of the element and the total power or energy risked in it. In the case of power lines, each segment is an independent feature in the resulting layer.

New layers are generated as temporary layers. They should be saved in the project's folder to make them permanent.

Finally, the .csv file includes more detailed information. The user can check the amount of power or energy risked in each element because of each vulnerability. This information might be helpful to assess the introduction of mitigation action in the system.

1.6 Case Study: 220 kV transmission network

GitHub repository includes a .zip file with a simple and customizable case study to introduce users to the tools.

The .zip file includes the following elements:

- 01. Hazards
 - WD.shp ► wind gusts hazard shapefile ¹
 - WF.shp ► wildfire gusts hazard shapefile ¹
- 02. PowerLines_Buses
 - buses_REE_ENDESA_lleida_220kV_nord_test.shp ► buses shapefile without consumption and generation data.
 - powerlines_REE_ENDESA_lleida_220kV_nord_test.shp ► power lines shapefile.
 - power_demand-generation.xlsx ► spreadsheet to adjust consumption and generation data.
- 03. Vulnerabilities_templates
 - Vulnerabilites_buses_template.xlsx ► Template to characterize each bus's vulnerability level and impact.
 - Vulnerabilites_powerlines_template.xlsx ► Template to characterize each power line's vulnerability level and impact.

1.6.1 System description

The provided example contains a 220 kV transmission network located in northern Spain. The system is composed of 24 buses (14 referred to main substations, 2 buses connected to the main grid and 8 additional nodes), and 19 overhead power lines.

¹See [1] for layer building process description.

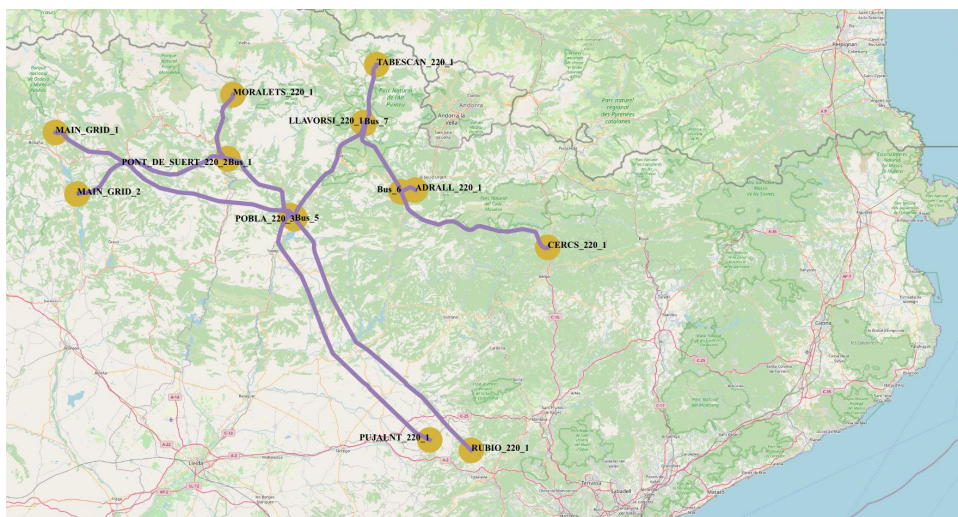


Figure 4: Buses (yellow) and power lines (purple) layers provided in the example case study.

The bus's location and name are obtained from OpenStreetMap [2] and Red Eléctrica [3]. The buses layer (buses_REE_ENDESA_lleida_220kV_nord_test.shp) does not include information about consumption/generation. This data should be generated and uploaded using the spreadsheet file (power_demand-generation.xlsx) following the instructions detailed in Section 1.6.2.

The power lines path and number of circuits are obtained from OpenStreetMap [2]. As for their technical characteristics, all have been defined as per the 490-AL1/64-ST1A 220.0 pandapower's standard type [4]. All power lines' required data is included in the provided layer (powerlines_REE_ENDESA_lleida_220kV_nord_test.shp).

1.6.2 Step-by-Step Guide to Run the Example

Follow this section to use the case study:

- **HAZARDS DATABASE**

1. Create in QGIS a group of layers called "*Hazards*"²
2. Upload the hazards layers WD.shp and WF.shp as: *Layer ► Add Layer ► Add Vector Layer ► Filter all files included in the folder and select only the .shp (WD.shp and WF.shp).*³
3. Verify that layers are included in "*Hazards*" group.
4. Once uploaded, the two layer might be combined to create a new layer → Follow Section 1.3

- **CASE STUDY**

1. Create in QGIS a group of layers called "*System_TBA*"²

²Group's name is essential to run the PlugIn.

³Note that all uploaded layers are in the same Reference Coordinate System [EPSG:25831 - ETRS89 / UTM zone 31N]

2. Upload the hazards layers buses_REE.ENDESA.lleida.220kV_nord_test.shp and powerlines_REE.ENDESA.lleida.220kV_nord_test.shp as: *Layer ► Add Layer ► Add Vector Layer ► Filter all files included in the folder and select only the .shp (buses_REE.ENDESA.lleida.220kV_nord_test.shp and powerlines_REE.ENDESA.lleida.220kV_nord_test.shp).*³
3. Verify that layers are included in "System_TBA" group.
4. Customize the generation and consumption profiles in the spreadsheet power_demand-generation.xlsx.
→ green cells should be filled.
 - **cons_gen_avg_mw** tab includes the instantaneous consumption and generation per season.^{4 5}
 - **cons_24h_mw** and **gen_24h_mw** include the daily profile per season.^{4 5}
 - **data** tab summarizes all data introduced in previous tabs. Data is shown in kW in this tab. In this tab, the user should specify the generation (controllable - PV or non-controllable - PQ).
5. Once filled, save **data** tab as .csv.
6. Upload the .csv in QGIS as *Add Layer ► Add Vector Layer ► Delimited Text ► Select the .csv file ► In file format select the option CSV (comma separated values) ► In geometry definite select the option No geometry (attribute only table)*
7. In the QGIS toolbox, select the **join attributes by field value** tool ► In **Input layer** select the buses layer previously uploaded in "System_TBA" group ► In **Table field** select "NAME" ► In **Input layer 2** select the attribute table that includes all .csv data ► In **Table field 2** select "NAME" ► In **Layer 2 field to copy** select all except "NAME" and "Voltage_kV" ► Press **Run** → A new buses layer will be generated including all consumption and generation information.
8. Calculate each element's impact under the created scenario following the instructions detailed in Section 1.4.

• RESILIENCE ANALYSIS

1. Fill the vulnerability templates (Vulnerabilites_buses_template.xlsx and Vulnerabilites_powerlines_template.xlsx) → green cells should be filled. Columns "_damage" should be filled considering the results obtained from the impact analysis.
2. Run the resilience analysis following the instructions detailed in Section 1.5

⁴Note that in this tab, data should be introduced in MW.

⁵Follow Section 1.4 Fields names

References

- [1] P. Muñoz Peña, M.-P. Montserrat, M. Cheah-Mane, and O. Gomis-Bellmunt, “Report describing the results of the case studies conducted,” Dec. 2024. [Online]. Available: <https://doi.org/10.5281/zenodo.14961741>
- [2] OpenStreetMap contributors, “Planet dump retrieved from <https://planet.osm.org>,” <https://www.openstreetmap.org>, 2017.
- [3] R. E. Española, “Conoce la capacidad de acceso,” 2025, accedido: 2025-04-28. [Online]. Available: <https://www.ree.es/es/clientes/generador/acceso-conexion/conoce-la-capacidad-de-acceso>
- [4] L. Thurner, A. Scheidler, F. Schäfer, J. Menke, J. Dollichon, F. Meier, S. Meinecke, and M. Braun, “pandapower — an open-source python tool for convenient modeling, analysis, and optimization of electric power systems,” *IEEE Transactions on Power Systems*, vol. 33, no. 6, pp. 6510–6521, Nov 2018.



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